Homework Week 4-5

Modern Physics

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For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

GENERAL QUANTUM MECHANICS PROBLEMS

- **Problems from the book**: 6.34 (use the identities in appendix 3C), 6.45, 6.46

- **Wavefunction normalization and Heisenberg uncertainty**
  A particle on the $[-1, 1]$ interval has the wavefunction
  $$\psi(x) = A(1 - x^2).$$
  Find the normalization constant A of the wavefunction. Calculate the position and momentum uncertainties and check that the Heisenberg relation is satisfied.

- **Harmonic oscillations of a N atom / Ex. 6.35**
  A nitrogen atom of mass $m_{\mathrm{N}} = 2.32 \times 10^{-26}\,\mathrm{kg}$ oscillates in one dimension at a frequency of $f = 1 \times 10^{13}\,\mathrm{Hz}$. What are its effective force constant κ and quantized energy levels E_n?

- **Tunneling**
  In a particular semiconductor device, electrons that are accelerated through a bias voltage $\Delta V = 3\,\mathrm{V}$ attempt to tunnel through a barrier of width $L = 0.7\,\mathrm{nm}$ and height $V_0 = 7.5\,\mathrm{eV}$. The potential is zero outside the barrier. (i) Verify that the system is in the *wide barrier limit*, and (ii) calculate the fraction of electrons that tunnel through the barrier using the approximate expression
  $$T = 16\,\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right)e^{-2\kappa L}.$$

- **Scattering on an interface**
  Consider a particle coming from $-\infty$ with energy $E > 0$ scattering on the potential step
  $$V(x) = \begin{cases} 0, & \text{if } x < 0, \\ V_0 & \text{if } x > 0, \end{cases}$$
  where $V_0 < E$ (possibly negative). This is a model for an electron crossing the interface between two different materials. Compute the transmission and reflection probabilities using the following steps:

  (a) Calculate the de Broglie wavelength λ_{L} of the electron on the left before it hits the step. What is the de Broglie wavelength λ_{R} on the other side of the step? Calculate the corresponding wavenumbers k_{L} and k_{R}.

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(b) Use the following trial wavefunction:

$$\psi(x) = \begin{cases} \psi_L(x) = Ae^{ik_L x} + Be^{-ik_L x}, & \text{if } x < 0, \\ \psi_R(x) = Ce^{ik_R x} & \text{if } x > 0, \end{cases}$$

where A, B and C are the complex amplitudes of the incoming, reflected, and transmitted plane waves respectively. (As A is arbitrary, you can set $A = 1$ in the following to simplify the calculation.) The wavefunction needs to be continuous at the interface, meaning $\psi_L(0) = \psi_R(0)$. Write down an equation relating the amplitudes using this.

(c) The derivative of the wavefunction also needs to be continuous, otherwise the second derivative wouldn't give a finite result,

$$\frac{d\psi_L}{dx}(0) = \frac{d\psi_R}{dx}(0).$$

Write down a second equation between the amplitudes using this, and express B/A and C/A with k_L and k_R.

(d) Determine the reflection coefficient using $R = |B/A|^2$. Sketch B/A and R as a function of V_0/E. What happens in the $V_0 = E$, $V_0 = 0$ and $V_0 \to -\infty$ limits?

(e) Determine the transmission coefficient T from R. The result is not equal to $|C/A|^2$, because the transmission probability is defined as the ratio of the incoming and outgoing probability currents, which depend on the velocity of the electrons that are unequal for incoming and outgoing in this case. Write down the correct expression in terms of C/A, k_L and k_R, and check that $T + R = 1$.

FROM 1D TO 2D AND 3D IN PROBLEMS WITH SEPARATION OF VARIABLES

Some of the one-dimensional (1D) problems discussed so far can be easily generalized to two and three spatial dimensions (2D or 3D). If the spatial variables x, y, ad z are *independent*, i.e., the potential is *separable*

$$V(x, y, z) = V(x) + V(y) + V(z),$$

the Schrödinger equation for a particle of mass m can be separated into a set of independent equations, one for each variable, that describe effective 1D problems

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right]\phi_{n_x}(x) = E_{n_x}\phi_{n_x}(x)$$

where E_{n_x} is the energy and $\phi_{n_x}(x)$ the corresponding eigenstate of the 1D problem. One also gets similar equations for the other spatial directions.
Then, the solution of the Schrödinger equation is given by

$$\psi_{n_x,n_y,n_z}(x, y, z) = \phi_{n_x}(x)\,\phi_{n_y}(y)\,\phi_{n_z}(z)$$

and the corresponding energy is given by

$$E_{n_x,n_y,n_z} = E_{n_x} + E_{n_y} + E_{n_z}.$$

In the following, we consider a few cases for familiar problems.

- **2D infinite well**
  Consider electrons in an infinite well in two spatial dimensions, with potential given by

  $$V(x, y) = \begin{cases} 0, & \text{if } 0 < x < L_x \text{ and } 0 < y < L_y, \\ \infty & \text{otherwise} \end{cases}$$

  Find: (i) the form of the normalized solution $\psi_{n_x,n_y}(x, y)$, (ii) the general expression of the corresponding energy E_{n_x,n_y}, (iii) the degeneracy (number of states with the same energy) of the first few excited states, e.g., for $n = n_x + n_y \leq 4$, for $L = L_x = L_y = 1\,\mathrm{nm}$.

- **3D infinite well**
  Consider electrons in an infinite well in three spatial dimensions, with potential given by

$$V(x,y,z) = \begin{cases} 0, & \text{if } 0 < x < L_x \text{ and } 0 < y < L_y \text{ and } 0 < z < L_z \\ \infty & \text{otherwise} \end{cases}$$

  Find: (i) the form of the normalized solution $\psi_{n_x,n_y,n_z}(x,y,z)$, (ii) the general expression of the corresponding energy E_{n_x,n_y,n_z}, (iii) the degeneracy of the first few excited states, e.g., for $n = n_x + n_y + n_z \leq 6$ for $L = L_x = L_y = L_z = 1\,\text{nm}$.

- **2D harmonic oscillator**
  Consider electrons in a harmonic potential in two spatial dimensions, with potential given by

$$V(x,y) = \frac{\kappa}{2}(x^2 + y^2).$$

  Find: (i) the form of the normalized solution $\psi_{n_x,n_y}(x,y)$, (ii) the general expression of the corresponding energy E_{n_x,n_y}, (iii) the degeneracy (number of states with the same energy) of the first few excited states, e.g., for $n = n_x + n_y \leq 4$.

- **3D harmonic oscillator**
  Consider electrons in a harmonic potential in three spatial dimensions, with potential given by

$$V(x,y,z) = \frac{\kappa}{2}(x^2 + y^2 + z^2).$$

  Find: (i) the form of the normalized solution $\psi_{n_x,n_y,n_z}(x,y,z)$, (ii) the general expression of the corresponding energy E_{n_x,n_y,n_z}, (iii) the degeneracy (number of states with the same energy) of the first few excited states, e.g., for $n = n_x + n_y + n_z \leq 6$.

- How do the above results change if the spring constants are unequal for various directions, i.e.

$$V = \frac{1}{2}(\kappa_x x^2 + \kappa_y y^2 + \kappa_z z^2)$$

EXTRA QUANTUM MECHANICS PROBLEMS

These are more advanced problems that are above the level we expect on the midterms, but are very useful to solve as practice. If you have any questions, feel free to contact the instructors.

Several electrons in a box

According to Pauli's principle, there can only be one electron in any given state. If we put several electrons in a box, they will fill the available energy levels in order, starting from the lowest one. (This neglects the Coulomb repulsion between electrons, we treat them as non-interacting particles for now). Every level can hold at most two electrons, one spin up, and one spin down, and the energy is independent of the spin.

- **Several electrons in a 1D box**
  - Write down the total energies corresponding to $N = 1, 2, .., 10$ electrons in a 1D box of length L in units of $E_0 = \frac{\hbar^2}{8mL^2}$. Be careful with correct counting including spin!
  - What is the total energy of $N = 1000$ electrons in the box?
    Hint: You can use the approximation for large N:
    $$\sum_{n=1}^{N} n^2 \approx \int_0^N n^2 dn = N^3/3.$$
- **Several electrons in a 2D box**
  - Write down the total energies corresponding to $N = 1, 2, .., 10$ electrons in a 2D box of length $L_x = L_y = L$ in units of $E_0 = \frac{\hbar^2}{8mL^2}$. Be careful with correct counting including spin!
  - What is the total energy of $N = 1000$ electrons in the box?
    Hint: As we already saw, the energy of a state is proportional to $k^2 = n_x^2 + n_y^2$. This is the equation of a circle of radius k in the $n_x - n_y$ plane, the lowest energy states are in the quarter circle given by $0 < n_x$, $0 < n_y$ and $k_{\max}^2 > n_x^2 + n_y^2$. For large $k_{\max}$, the number of states in this quarter circle is approximately the same as its area times 2 (because of spin). Using this, find $k_{\max}$.
    Next, we find the total energy of these states. You can use the approximation for large $k_{\max}$:
    $$\sum_{0<n_x,n_y}^{n_x^2+n_y^2<k_{\max}^2} (n_x^2 + n_y^2) \approx \iint_{0<n_x,n_y}^{n_x^2+n_y^2<k_{\max}^2} (n_x^2 + n_y^2)\, dn_x dn_y = \int_0^{k_{\max}} dk\, k \int_0^{\pi/2} d\phi\, k^2.$$
- **Several electrons in a 3D box**
  Same as the above, but for a 3D box of length $L_x = L_y = L_z = L$.
- **Force of an electron in a box**
  - An electron of mass m is in its ground state in a box of length L. Calculate the force the electron exerts on the wall of the box.
    Hint: Assuming that the wall of the box is movable, calculate the energy E of the electron as a function of the length of the box. The force is given by $F = -\frac{dE}{dL}$.
  - Assume that the wall of the box is movable, and is held in place by a spring that exerts a force $F_{\text{spring}} = -kL$. Calculate the equilibrium length of the box.
  - Repeat the calculation with a large number N electrons in the box.
- **Scattering on a Dirac delta potential**
  Consider a particle coming from $-\infty$ with energy $E > 0$ scattering on the Dirac delta potential barrier
  $$V(x) = \alpha\, \delta(x)$$
  where α is a constant. Compute the transmission and reflection probabilities using the following steps:
  (a) Calculate the de Broglie wavelength λ and the corresponding wavenumber k for the electron away from $x = 0$

(b) Use the following trial wavefunction:

$$\psi(x) = \begin{cases} \psi_L(x) = Ae^{ikx} + Be^{-ikx}, & \text{if } x < 0, \\ \psi_R(x) = Ce^{ikx} & \text{if } x > 0, \end{cases}$$

where A, B and C are the complex amplitudes of the incoming, reflected, and transmitted plane waves respectively. (As A is arbitrary, you can set $A = 1$ in the following to simplify the calculation.) The wavefunction needs to be continuous everywhere, meaning $\psi_L(0) = \psi_R(0)$. Write down an equation relating the amplitudes using this.

(c) If a function $f(x)$ has a discontinuity (jump) at $x = 0$, such that

$$f(x) = \begin{cases} f_L(x) & \text{if } x < 0, \\ f_R(x) & \text{if } x > 0, \end{cases}$$

then its derivative is expressible with a Driac delta as

$$f'(x) = (f_R(0) - f_L(0))\delta(x) + \begin{cases} f'_L(x) & \text{if } x < 0, \\ f'_R(x) & \text{if } x > 0. \end{cases}$$

Substituting ψ into the Schrödinger equation and using this result, match the jump in the wavefunction's derivative to the delta potential, write down a second equation between the amplitudes, and express B/A and C/A with k and α.

(d) Determine the reflection coefficient using $R = |B/A|^2$. Sketch B/A and R as a function of E/α.

(e) Determine the transmission coefficient using $T = |C/A|^2$, and check that $T + R = 1$.

- **Kronig-Penney model**

Consider the potential

$$V(x) = -\gamma \sum_{j=-\infty}^{\infty} \delta(x - aj)$$

where $\gamma > 0$ and $a > 0$ are constants. This is a simple model of an electron moving in an infinite crystal with atoms placed at distance a from each other.

As usual, we want to find the eigenstates and the corresponding eigenvalues (energies). Since this model has an infinite amount of 'sites' (the Dirac peaks) that a particle can interact with, there will be many eigenstates and corresponding energies. Find the energy region (band) for which eigenstates are possible.

Hint: Between given Dirac-delta sites $j - 1$ and j, you should search for the wavefunction in the form

$$\psi_j(x) = A_j e^{\alpha(x-ja)} + B_j e^{-\alpha(x-ja)},$$

where, depending on the sign of the energy E, α is either real $\alpha \in \mathbb{R}$ or purely imaginary $\alpha = ik \in \mathbb{C}$, $k \in \mathbb{R}$. First find α as a function of the energy E using the Schrödinger equation between the sites.

Next use the wavefunction matching conditions at the site at position $x = ja = 0$ (see the previous problem) to express coefficients A_1 and B_1 in terms of A_0 and B_0. These are linear equations, rewrite them in a matrix form with a so called transfer matrix T:

$$T \begin{pmatrix} A_0 \\ B_0 \end{pmatrix} = \begin{pmatrix} A_1 \\ B_1 \end{pmatrix}$$

What are the eigenvalues of T?

Because of the translation invariance of the system, the same relation is valid at every site (you can check this explicitly):

$$T \begin{pmatrix} A_j \\ B_j \end{pmatrix} = \begin{pmatrix} A_{j+1} \\ B_{j+1} \end{pmatrix}.$$

As a result, knowing the amplitudes at $j = 0$, we can express the amplitudes for any j as

$$T^j \begin{pmatrix} A_0 \\ B_0 \end{pmatrix} = \begin{pmatrix} A_j \\ B_j \end{pmatrix}.$$

We want an eigenstate that doesn't grow exponentially in either direction, this is equivalent to requiring T to be a unitary matrix. It is easy to check that the determinant of T is 1, and its eigenvalues are roots of unity: $\lambda_{1,2} = e^{\pm i\varphi}$ and it can be seen that if $\mathrm{Tr}\, T = 2\cos\varphi \leq 2$ then T is guaranteed to be unitary.

From the $\mathrm{Tr}\, T \leq 2$ condition you should find the following constraint on the possible values of α (or $k = -i\alpha$ depending on the sign of E):

$$E < 0 :$$
$$\frac{m\gamma}{\hbar^2} a \geq \alpha a \tanh\left(\frac{\alpha a}{2}\right)$$
$$E > 0 :$$
$$\frac{m\gamma}{\hbar^2} a \leq k a \tan\left(\frac{k a}{2}\right)$$

Together with the relation between α and E found earlier, these inequalities give the allowed ranges of E for which eigenstates exist, these are called energy bands. Use graphical or numerical methods to solve these equations.